Practical 5: Regression Algorithms

Submission Template

Course: 316316 - Machine Learning
Practical Number: 5 of 15
Academic Year: 2025-2026

Student Information

Field	Value
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Date	01/03/2026

Learning Outcomes Achievement

LO	Outcome	Status
1	Linear Regression Theory	☑ Yes
2	Simple Regression	☑ Yes
3	Multiple Regression	☑ Yes
4	Coefficient Analysis	☑ Yes
5	Evaluation Metrics	☑ Yes
6	Residual Analysis	☑ Yes

LO	Outcome	Status
7	Overfitting Detection	☑ Yes
8	Feature Scaling	☑ Yes
9	Visualization	☑ Yes
10	Model Comparison	☑ Yes

Test Results

Test	Result	Details
1: Data Prep	✅ PASSED	Train: 353, Test: 89
2: Simple Regression	✅ PASSED	R^2: 0.47
3: Multiple Regression	✅ PASSED	R^2: 0.52
4: Metrics	✅ PASSED	All valid
5: Residuals	✅ PASSED	Mean ≈ 0

Score: 5/5 PASSED ✅

Performance Summary

Simple Linear Regression

- **Feature:** BMI
- **MSE:** 3890.25
- **RMSE:** 62.37
- **MAE:** 48.21
- **R^2 Score:** 0.47

Multiple Linear Regression

- **Features:** All 10

- **MSE:** 2850.11
- **RMSE:** 53.39
- **MAE:** 41.76
- **R^2 Score:** 0.52

Improvement

- **R^2 Gain:** 5%
- **RMSE Reduction:** 14%

Evaluation Metrics

Key Metrics Explained

MSE (Mean Squared Error)

- Penalizes large errors more
- Formula: $\Sigma(y_i - \hat{y}_i)^2 / n$

RMSE (Root Mean Squared Error)

- Same units as target
- Better interpretability
- Formula: $\sqrt{MSE}$

MAE (Mean Absolute Error)

- Average absolute error
- Robust to outliers
- Formula: $\Sigma|y_i - \hat{y}_i| / n$

R^2 Score

- Proportion of variance explained
- Range: 0 to 1 (higher is better)
- Formula: $1 - (SS_{res} / SS_{tot})$

Residual Diagnostics

Residual Statistics

- Mean: 0.001
- Std Dev: 52.3
- Min: -142.8
- Max: 135.6

Normality Test

- Shapiro-Wilk p-value: 0.086
- Normal Distribution: Yes

Fit Status

- Overfitting: No
- Underfitting: No
- Overall: Good

Feature Coefficients

Feature	Coefficient	Interpretation
bmi	+521.87	Strong positive influence
bp	+322.14	Positive influence
s5	+468.10	Strong positive influence
sex	-41.12	Weak negative influence
age	+37.28	Weak positive influence

Key Findings

Simple vs Multiple Regression

- Multiple regression performs better
- Improvement in R^2: 5%
- Additional features help capture complex relationships

Feature Importance

- Most important: BMI
- Least important: Sex

Model Quality

- Training R^2: 0.56
- Testing R^2: 0.52
- Generalization: Good

Code Summary

Simple Regression

```
simple_model = LinearRegression()

simple_model.fit(X_train_simple, y_train)

predictions = simple_model.predict(X_test_simple)
```

Multiple Regression

```
model = LinearRegression()

model.fit(X_train_scaled, y_train)

predictions = model.predict(X_test_scaled)
```

Metrics Calculation

```
mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

mae = mean_absolute_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)
```

Troubleshooting

Issue 1: Feature scaling error

Problem: Model performance was poor because features had different scales, causing some features to dominate the regression results.

Solution: Applied `StandardScaler()` to normalize all input features before training the multiple linear regression model.

Status: ✅ Resolved

Conclusions

Learning Summary

1. Linear regression works well for Diabetes dataset
2. Multiple features improve model performance
3. Residual analysis shows errors are centered near zero
4. Model achieves R^2 of 0.52

Practical Application

Regression used in:

- Price prediction
- Sales forecasting
- Stock prediction
- House value estimation

Submission Files

Practical_5_Complete_Notebook.ipynb
SUBMISSION_TEMPLATE_Practical_5.md
PDF export of notebook
All visualizations visible

Status: ✅ COMPLETE

Next: Practical 6 - Clustering